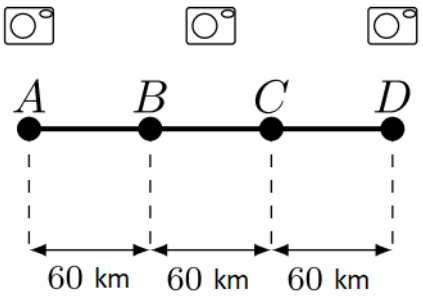
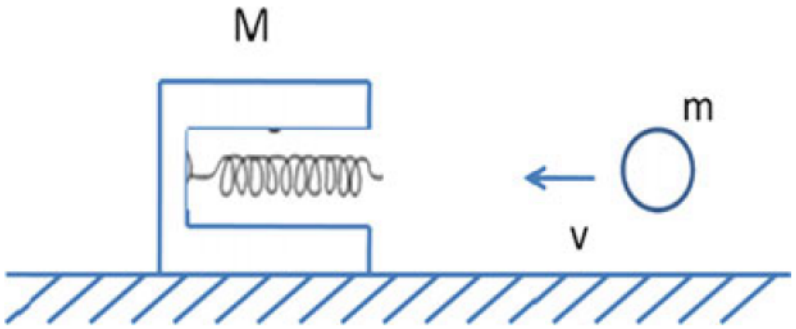
1. Mashina A punktdan D punktga B va C punktlardan o‘tib boradi. Qo‘shni punktlar orasi $60km$. Ruxsat etilgan eng yuqori tezlik AB oraliqda $60km/soat$, BC da - $90km/soat$, CD da - $120km/soat$. Yo‘lda mashina qaysi vaqtda o‘tganini qayd etuvchi kameralar o‘rnatilgan. Birinchi kamera A punktda, ikkinchisi B va C punktlar oralig‘idagi masofaning yarmida,uchinchisi D punktda. Mashina birinchi kameradan 13 : 20 da, ikinchisidan 14 : 30 da va uchinchisidan 15 : 30 da o‘tgan. Qaysi kameralar oralig‘idagi hududda mashina aniq tezlik qoidasini buzgan?

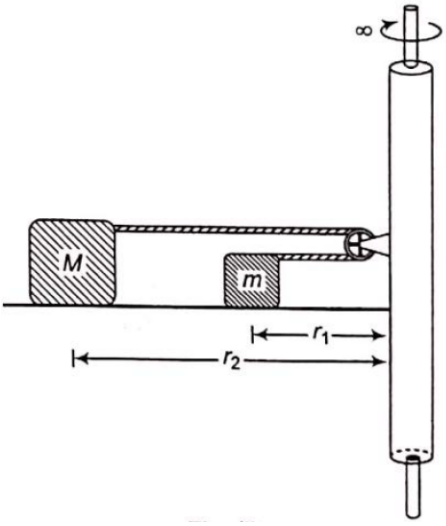


2.m massali sharcha gorizontal yo‘nalishda v tezlikda uchib kelmoqda. U gorizontal silliq stol ustida yotgan M massali brusokka mahkamlangan bkirligi k bo‘lgan prujinaga uriladi (rasmga qarang). To’qnashuvda brusok siljishini hisobga olib prujinaning maksimal deformatsiyasini toping.

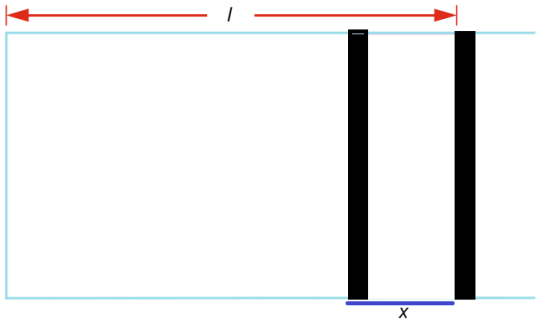


3. Massalari m va M bo‘lgan brusoklar bir biriga vaznsiz arqon bilan vertikal o‘q atrofida ω burchak tezlik bilan aylanuvchi sterjenga mahkamlangan, ishqalanishsiz aylanadigan va vaznsiz

blokdan o'tkazib bog'langan (rasmga qarang). Brusoklar va sirt orasidagi ishqalanish koeffitsientlari mos ravishda μ_1 va μ_2 bo'lsa, brusoklar radial siljiy boshlashi uchun ω ni toping($M > m$).



4.Gorizontal qo'zg'almas slindr ichidagi ideal gaz erkin siljiy oladigan m massali porshen bilan berkitilgan. Muvozanat holatida, ya'ni tashqi va ichki bosim P ga teng bo'lganda gaz slindrning l uzunligini egallaydi. Jarayon izotermik deb sistemaning kichik tebranishlar davrini toping.Porshen yuzasi A.



5. Maxsus masspektrometrda magnit maydon induksiyasi $0.20Tl$. Spektrometrda Uranning ikki izotoplarini ajratish vazifasi qo'yilgan: $^{235}_{92}U$ massasi $3.90 \cdot 10^{-25}kg$ va $^{238}_{92}U$ massasi $3.95 \cdot 10^{-25}kg$. Izotoplar $+e = 1.6 \cdot 10^{-19}c$ zaryadga ega bo'lib, spektrometrda ajralishi uchun trayektoriyalari egrilik radiuslari $2mm$ farq qilishi kerak. Bunga erishish uchun ionlar tezlatiladigan potensiallar farqini toping.